

Hands-on Lab: Travel Recommendation Web Application



Estimated time needed: 90 minutes

Introduction

In this final project, you will create a webpage layout for a travel recommendation platform.

This website will include:

- A navigation bar with links to the home page, "About Us" and "Contact Us" pages
- A search bar allowing users to input destinations or keywords to search for recommendations
- A content section that showcases an enticing background image, social media links, and a headline emphasizing exploration and destination discovery
- A brief description promoting cultural exploration
- A "Book Now" button

You will be able to implement the knowledge and skills you have gained from working on the practice project to help you handle dynamic functionalities, like search recommendations and result presentation.

Learning objectives

After completing this lab, you will be able to:

- **Enhanced user navigation:** Implement an intuitive navigation bar with clear links to the Home, About Us, and Contact Us sections, ensuring a user friendly website.
- **Search functionality integration:** Develop a search feature allowing users to input destinations or keywords, enabling the 'Search' and 'Clear' buttons to retrieve and reset dynamically displayed recommendations promptly.
- **Visual appeal and brand representation:** Create an aesthetically pleasing landing page with visually engaging elements, including a logo, social media icons, and captivating visuals, to enhance brand recognition and user engagement.
- **JavaScript functionality:** Utilize JavaScript to enable dynamic functionalities, such as live search recommendation fetching, interactivity for user inputs, or potential API integrations, enhancing the website's functionality and user experience.

Prerequisites

- Basic knowledge of HTML and GitHub
- Basic understanding of JavaScript functions, the Fetch API method, strings, arrays
- Web browser with a console (Chrome DevTools, Firefox Console, and so on)

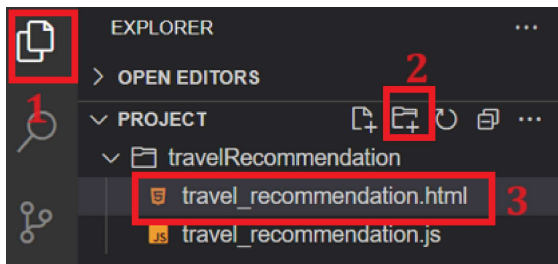
Setting up the environment

1. You need to create one blank online GitHub repository and name it according to the name of your travel recommendation website.

Important! Make sure your repository is public because your peers will need to review your work. Also, ensure that the name of the repository and the folder you are going to create in Point 3 should be the same.

2. Do not create README.md files or others not included in this project.

3. On the window to the left, click on **Explorer** as shown at **1** in the screenshot below. Click on the project folder and again click on the icon highlighted in red at **2** in the screenshot. Enter the folder name **travelRecommendation**. It will create a folder for you. Then select the **travelRecommendation** folder shown at **3**, right-click, and select **New File**. Enter the file name **travel_recommendation.html** and click **OK**. It will create your HTML file.



4. Again, right click on the **travelRecommendation** folder and select a **New File**. Enter the file name **travel_recommendation.js** and click **OK**. It will create your JavaScript file.
5. To create a JSON file, right-click on the **travelRecommendation** folder again and select a **New File**. Enter the file name **travel_recommendation_api.json** and click **OK**. It will create your JSON file.
6. Click on this link [travel_recommendation_api](#) and copy data, then paste into the **travel_recommendation_api.json** file and then save it.

Note: This data will act as API data where details for travel-based recommendation information are available.

Creation of Travel Recommendation Web Application

Note:

Please save your project and capture screenshots wherever necessary.

1. As mentioned in the **Final Project Overview Criteria**, you can submit your project deliverables through either Option 1: AI-Graded Submission and Evaluation or Option 2: Peer-Graded Submission and Evaluation.
2. **For Option 1: AI-Graded Submission and Evaluation:**
 - Submission requires the **public GitHub repository URL** or **public URL of the deployed website** for Task 1 to 4, **code snippet** for Task 5 and **screenshots** for Tasks 6 to 8.
3. **For Option 2: Peer-Graded Submission and Evaluation:**
 - Submission requires the **public GitHub repository URL of your Travel recommendation website** and **public repository URL of the deployed Travel recommendation website** for Tasks 1 and 2.

Task 1: HTML template

Create a basic HTML template structure in the **travel_recommendation.html** file. It should include these tags:

- `<html>`
- `<head>`
- `<body>`
- Within the `<head>` tag include a `<title>` tag.

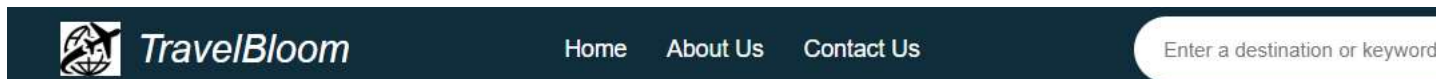
Write your website name in the `<title>` tag to set up the title of each web page.

Task 2: Navbar

For this task, you need to create a navigation bar. The navbar should contain the following:

- **Home** page link using anchor tag `<a>` to navigate to home page
- **About us** page link using anchor tag `<a>` to navigate to about us page
- **Contact Us** page link using anchor tag `<a>` to navigate to contact us page
- A **search bar** to allow users to enter keywords to search for recommendations
- A **Search** button to execute the search after the user clicks on it
- A **Reset** button to clear results.

The screenshot below shows a sample navigation bar.



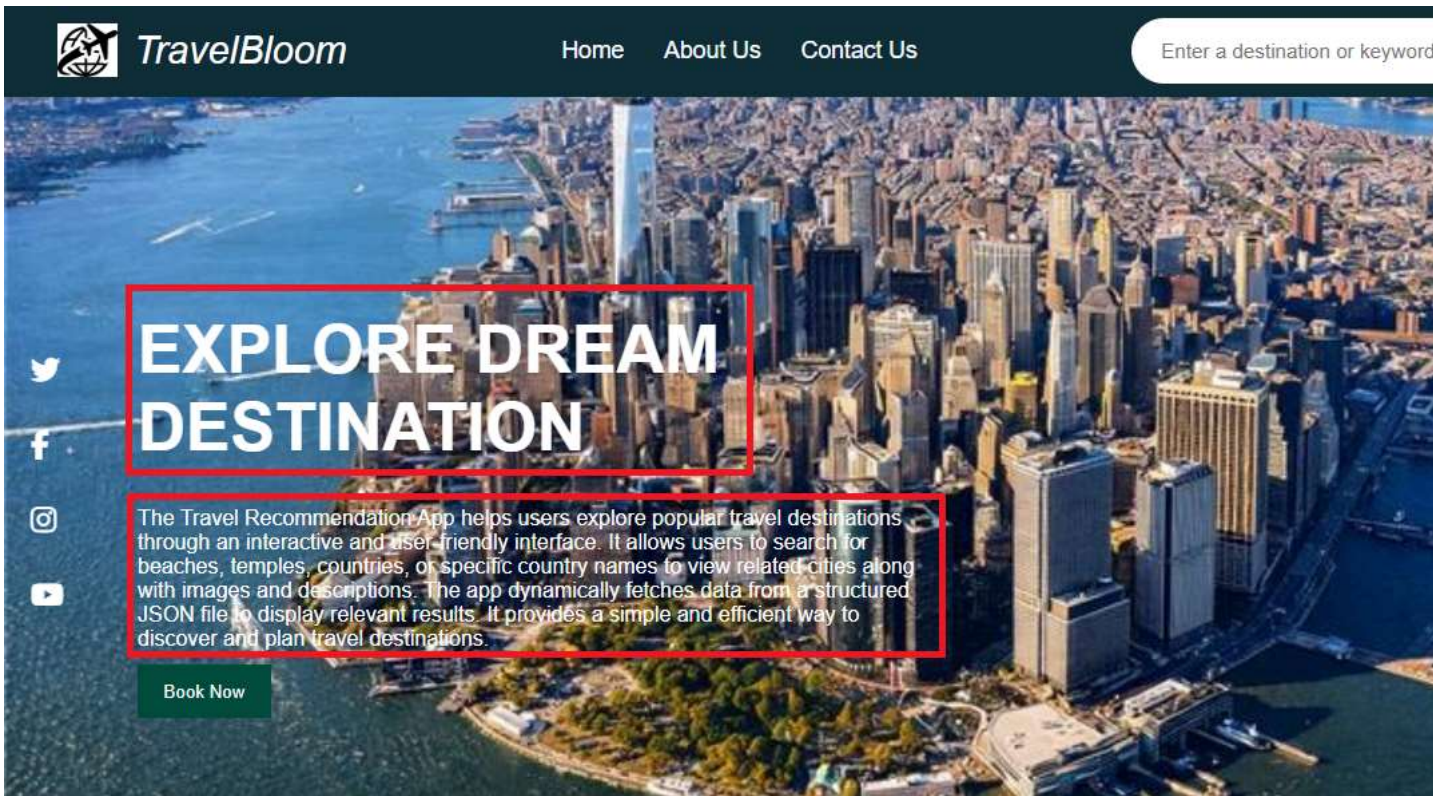
You can also include your website name and logo if you wish.

Task 3: Home page

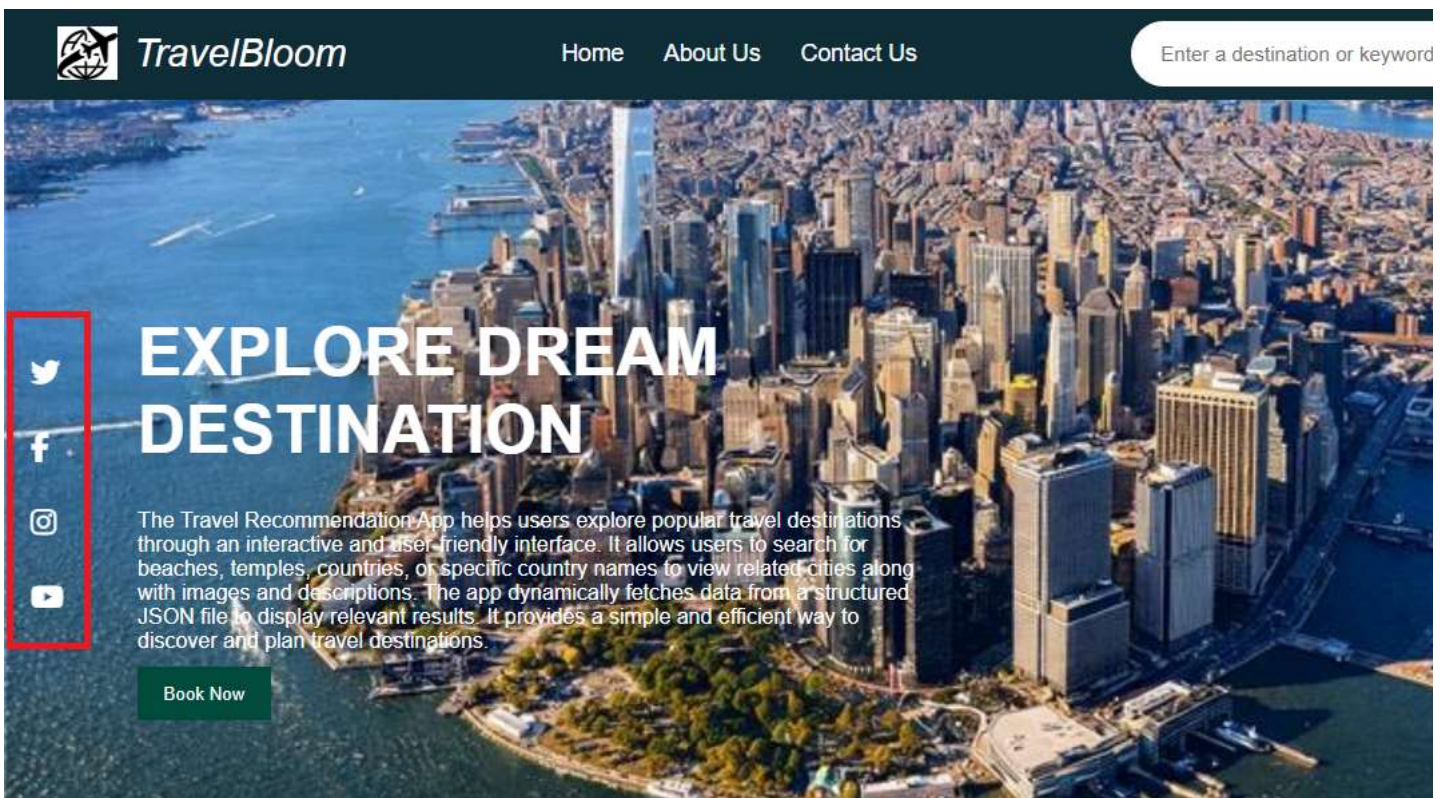
In this task, you will design a **Home** page which must include:

- A background image
- An introduction to this website

You can see an example home page with these elements in this screenshot:



You can also include icons on the home page for social media platforms.



Note: You can also include CSS in the HTML code based on the themes and the images you choose.

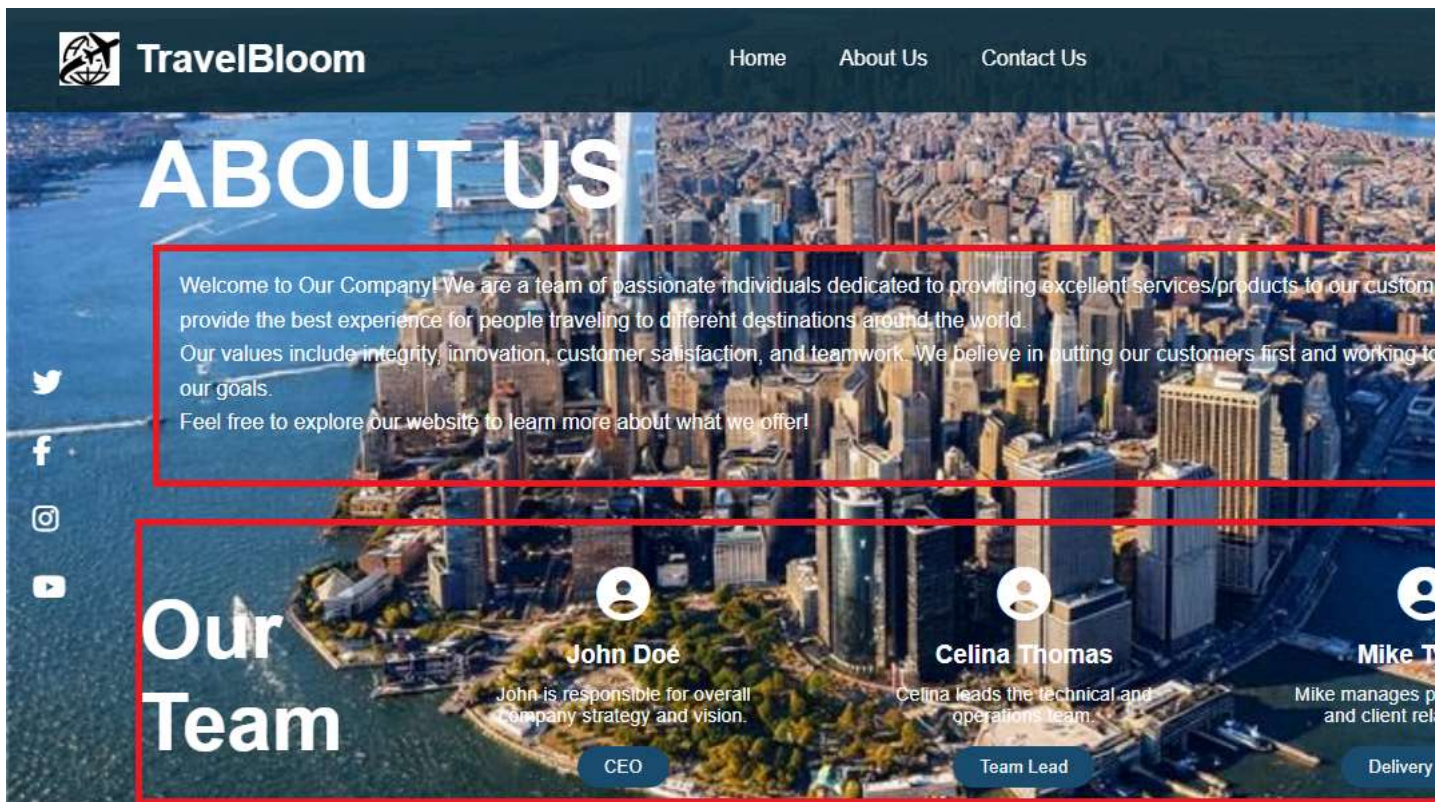
Task 4: About us

In this task, you need to create an **About Us** page, which should include:

- Information about the company
- An introduction to the team members, displaying their names and their designated roles

Note: Make sure that you include the code for the navbar. The navbar on this page should only include Home, About Us, and Contact Us menu items, *not* the search bar and buttons.

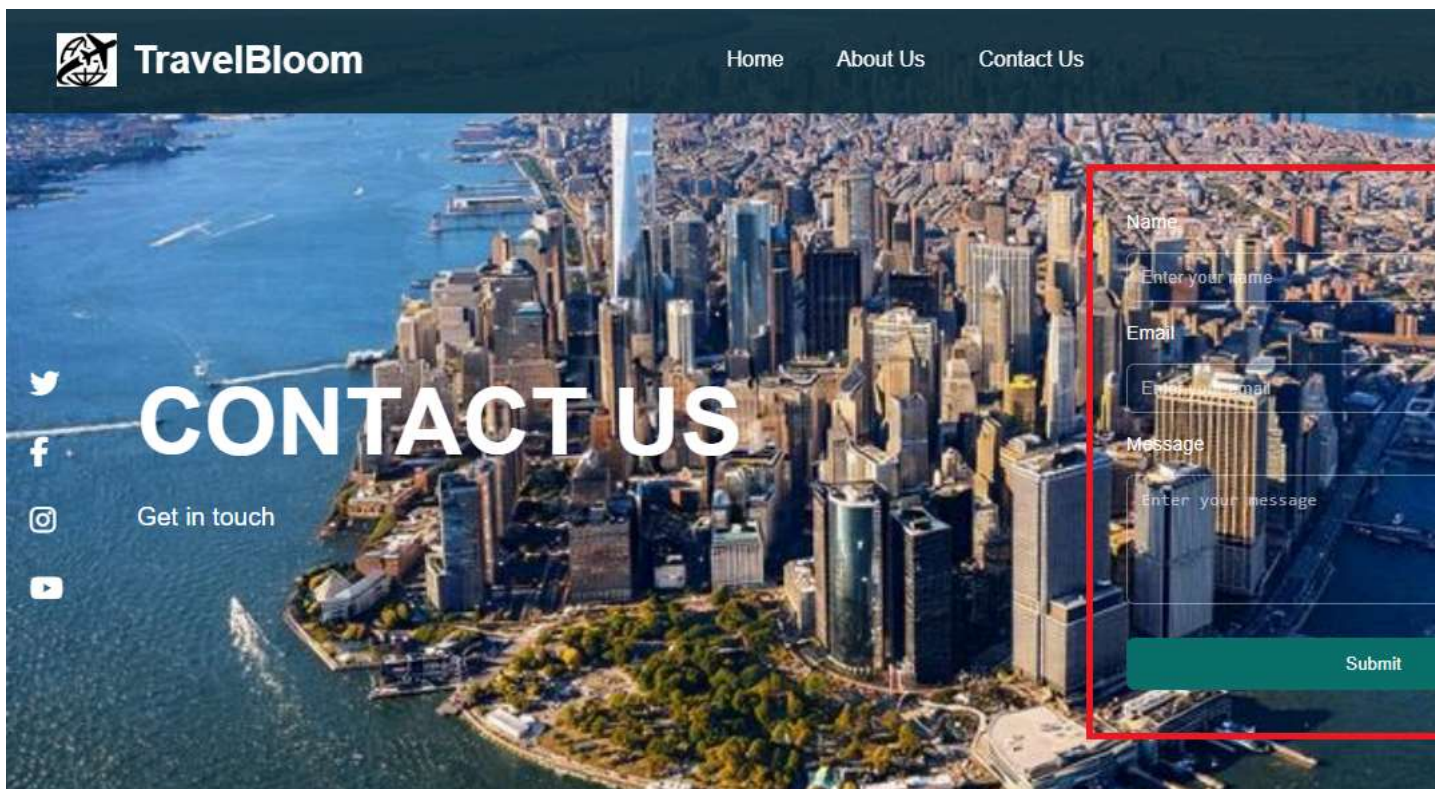
The screenshot below shows a sample **About Us** page.



Task 5: Contact us

In this task, you need to create a **Contact Us** page, which should contain the following:

- A form for users in case they want to reach out
- The form should contain:
 - `<input>` boxes for name and email
 - `<textarea>` for users to write their message
 - **Submit** button to submit the form



Task 6: Recommendation results

Now, you need to create logic in your JavaScript file to show results for your recommendations.

Note: You should check the output of your code while developing your JavaScript. Directions to view your output are on the next page of these instructions.

- Fetch data from the `travel_recommendation_api.json` file using the `fetch` API method, from there you can fetch travel-related details, such as the name of the place. You need to have your own images for every `imageUrl` in the JSON file.

```

{id": 1,
"name": "Australia",
"cities": [
  {
    "name": "Sydney, Australia",
    "imageUrl": "enter_your_image_for_sydney.jpg",
    "description": "A vibrant city known for its iconic landmarks like
  },

```

Use image according to your refer

- Fetch the data from the JSON using the `fetch API()` method. To check if you can access the data, you use `console.log` to see if the result is displayed.
- It is good if the `console.log` logs the data. Otherwise, you need to look for a different API.

Task 7: Keyword searches

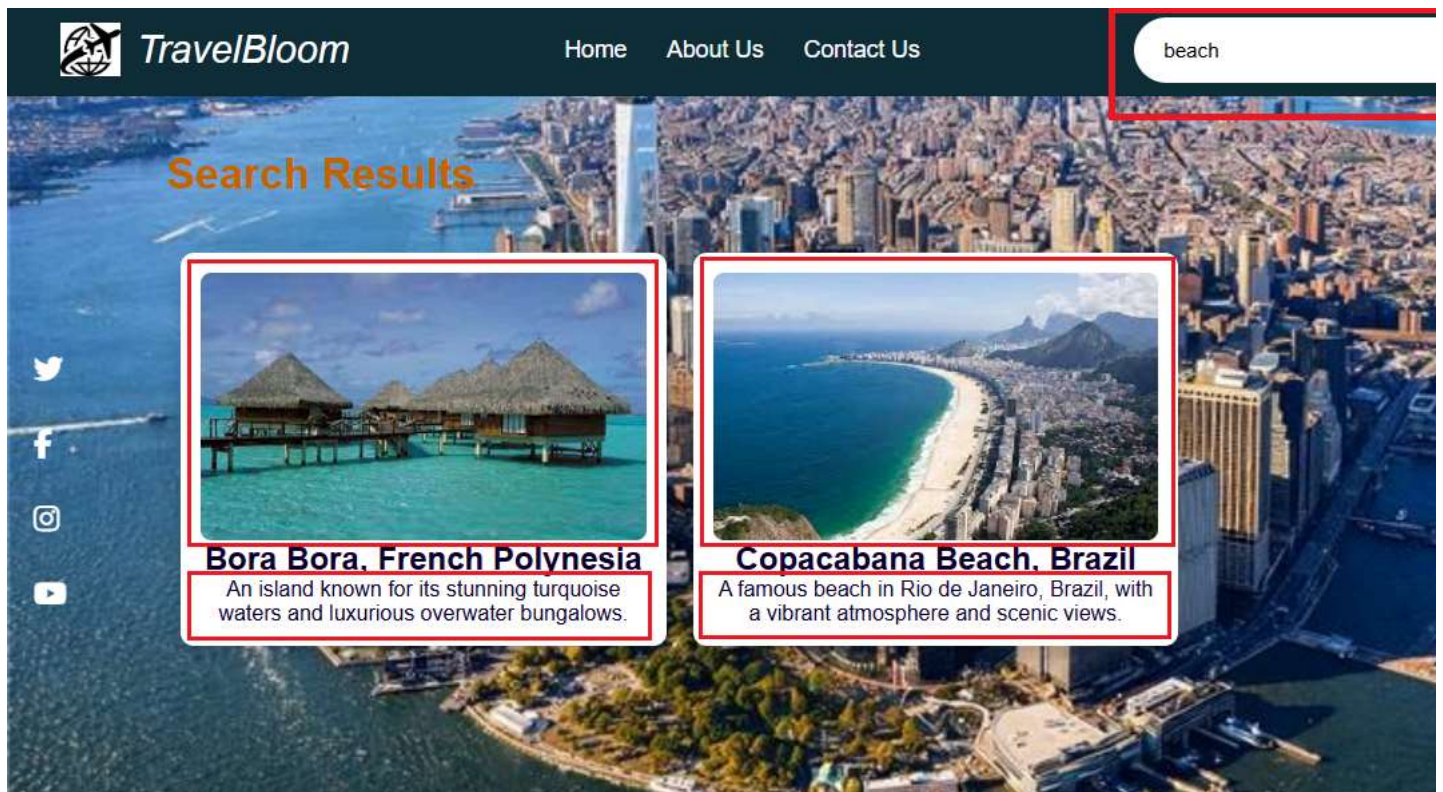
In this task, you will write JavaScript to accept these keywords and variations the user will enter in the search field in your navigation bar on the home page.

- For example, if the user enters "beach," or "beaches," "Beach" or "BEACH," then you need to write JavaScript code so that it accepts all variations of this keyword.
- For uppercase letters in the keyword, you can convert them to lowercase in your JavaScript using the string manipulation `toLowerCase()` method.
- Similarly, you need to create logic to match keywords entered for **temples** and **countries**.
- The website should display results only after the user clicks the **Search** button.

Task 8: Recommendations

In this task, you need to fetch the details of the places you recommend based on which keyword the user enters: *beach*, *temple*, or *country*.

For each of these three keywords, your results should display at least two recommendations, an image, and a description. Example screenshot is shown below.



Note: Use **display: grid** for the results section so that all images appear properly aligned in the same row on a single page. This layout will ensure that all result images are clearly visible and can be used for final evaluation.

Task 9: Clear button

Create logic in your JavaScript file for a **clear** button to clear the results. To implement this feature, you can create a function that will be called after clicking on the **clear** button in the navbar.

Task 10: Country date and time (optional)

In this optional task, you can create logic in your JavaScript to display the time in the country you recommend.

For Example:

```
const options = { timeZone: 'America/New_York', hour12: true, hour: 'numeric', minute: 'numeric', second: 'numeric' };
const newYorkTime = new Date().toLocaleTimeString('en-US', options);
console.log("Current time in New York:", newYorkTime);
```

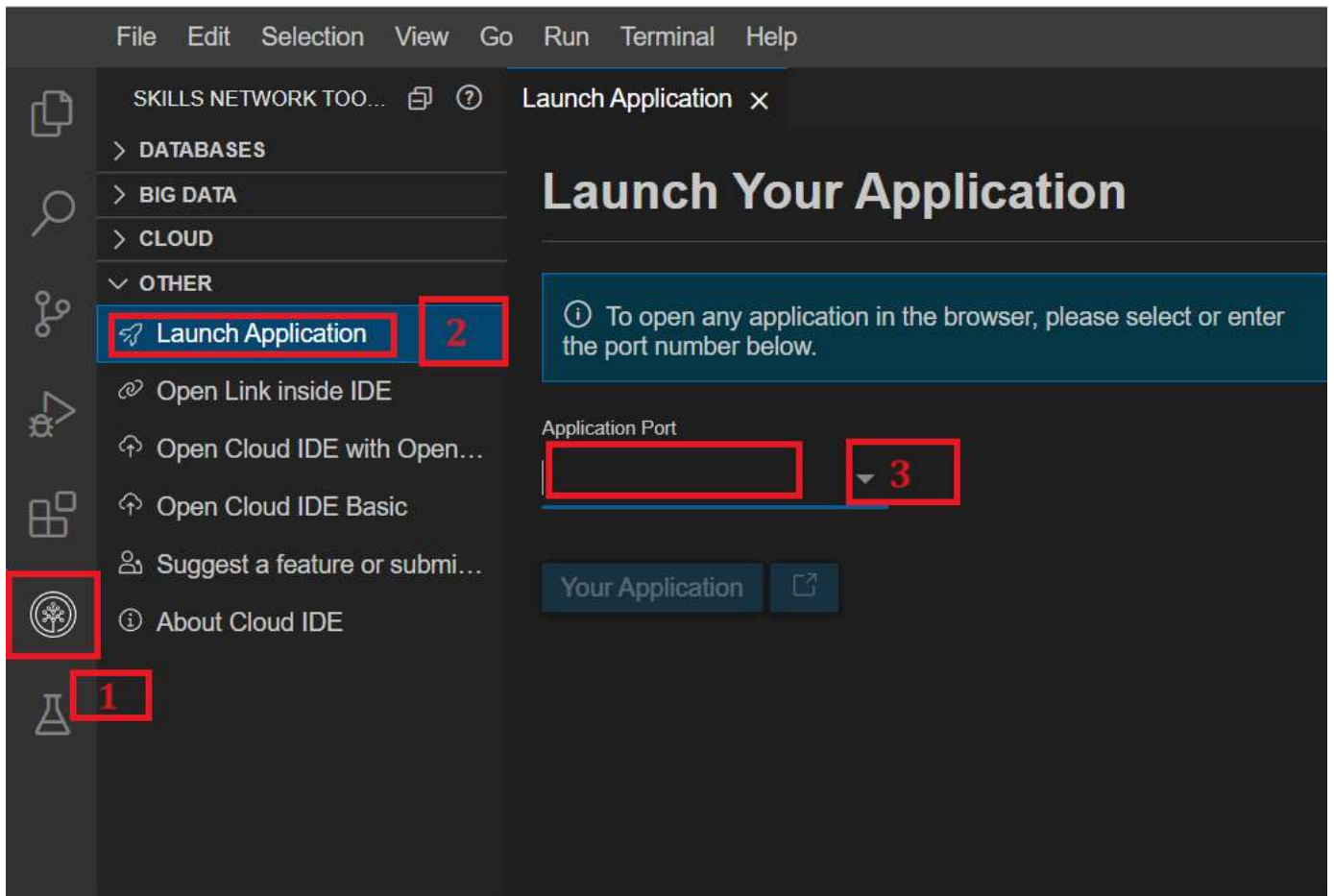
- This JavaScript code retrieves the current time in the New York time zone using the Date object and the toLocaleTimeString method.
 - `const options = { timeZone: 'America/New_York', hour12: true, hour: 'numeric', minute: 'numeric', second: 'numeric' };`
 - `timeZone: 'America/New_York'` sets the time zone to New York. `hour12: true` specifies that the time should be displayed in 12-hour format (AM/PM).
 - `hour: 'numeric', minute: 'numeric', and second: 'numeric'` indicate that the hour, minute, and second components of the time should be displayed numerically.
 - `const newYorkTime = new Date().toLocaleTimeString('en-US', options);`
 - `new Date()` creates a new Date object representing the current date and time.
 - `toLocaleTimeString('en-US', options)` formats the time according to the specified options in the 'en-US' locale (English - United States) and the provided options object (options). This method returns a string representing the time in the specified format.
 - `console.log` outputs the string "Current time in New York:" along with the formatted time retrieved in the newYorkTime variable.

Check the output

1. To view your HTML page in your browser, use the built-in Live Server extension. Select the file **travel_recommendation.html** within the project folder and right-click on it. Choose **Open with Live Server**.
2. A notification will appear at the bottom right, indicating that the server started on port 5500.



3. Click on the **Skills Network** icon button on the left, shown at 1 in the screenshot below. It will open the "Skills Network Toolbox" menu. Click the Other, then Launch Application shown at 2 in the screenshot below. From there, enter the port number 5500 shown at 3.



4. Click on this button 

5. Your default browser will open. You will see the **travelRecommendation** folder. Click on that folder name and then click again on the **travel_recommendation.html** file.

6. It will open the front page, and you can see the output.

Perform git commands

1. You need to save all your files to perform git commands.
2. Perform `git add`, `git commit`, and `git push` commands to update changes inside the **travelRecommendation** folder inside your GitHub repository that you created at the start for proper code management.

Note: If you're pushing code without cloning from a GitHub repository, remember to utilize "origin2" for the `git push` command. If you've logged out and want to resume work on the project, clone the repository from the GitHub where you previously pushed the code. After cloning, you can simply use `git push origin` without the need to execute `git remote add...` to push your changes directly.

Create git pages

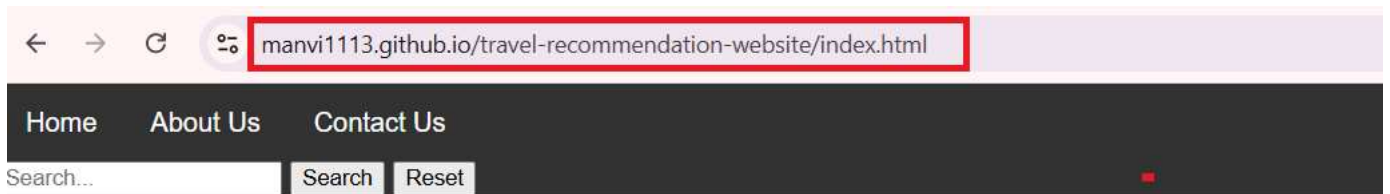
Create git pages for your website as practiced in the **Practice Project**.

Important! Make sure that your repository is public and that you have saved the URL for both the GitHub repository and for the link, which you will generate after creating Git pages for your website for graded review.

Instructions to Verify Your Site on GitHub Pages

If you are using `index.html` file:

1. Open your browser and navigate to the URL:
`https://<your-username>.github.io/<your-repo-name>/`
2. Verify that the site loads correctly and displays the content from your `index.html` file and automatically loads the homepage of the website.



Welcome to Travel Re

Explore amazing destinations and p

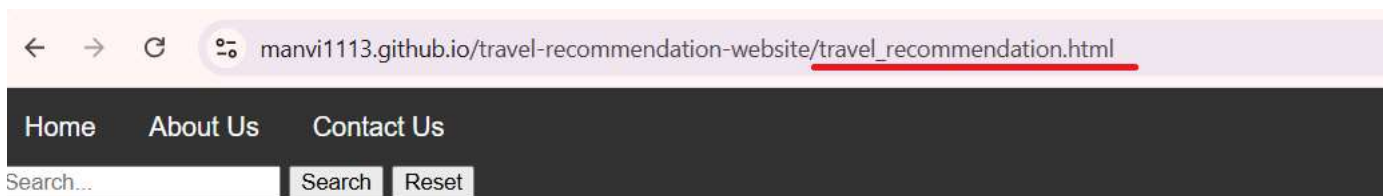
[Facebook](#) [Twitter](#) [Insta](#)

If you are using travel_recommendation.html file:

1. Open your browser and navigate to the URL and append the name of the html file you are using, it will automatically load the homepafge for you.

`https://<your-username>.github.io/<your-repo-name>/travel_recommendation.html`

2. Check that the travel_recommendation.html file loads correctly and displays the expected content.



Welcome to Travel Re

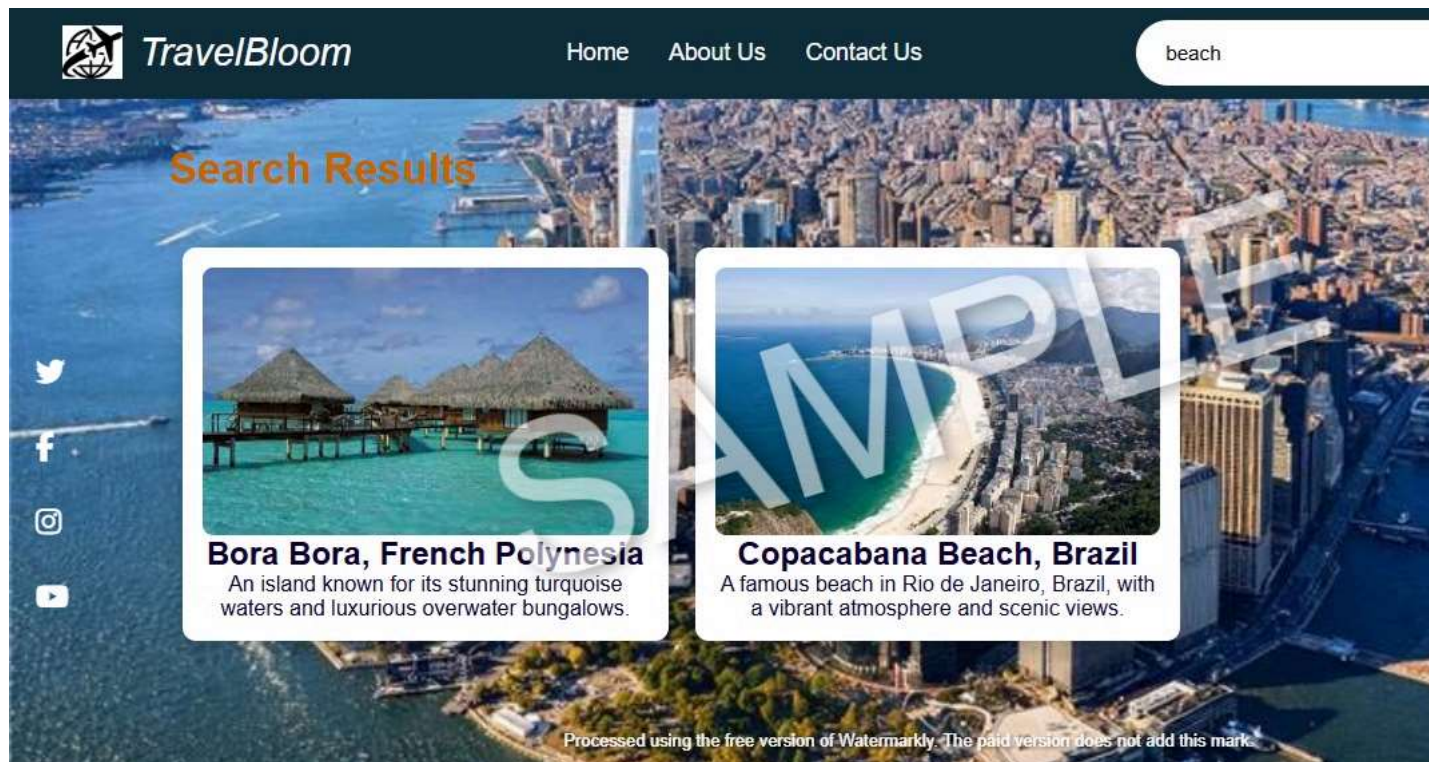
Explore amazing destinations and p

[Facebook](#) [Twitter](#) [Insta](#)

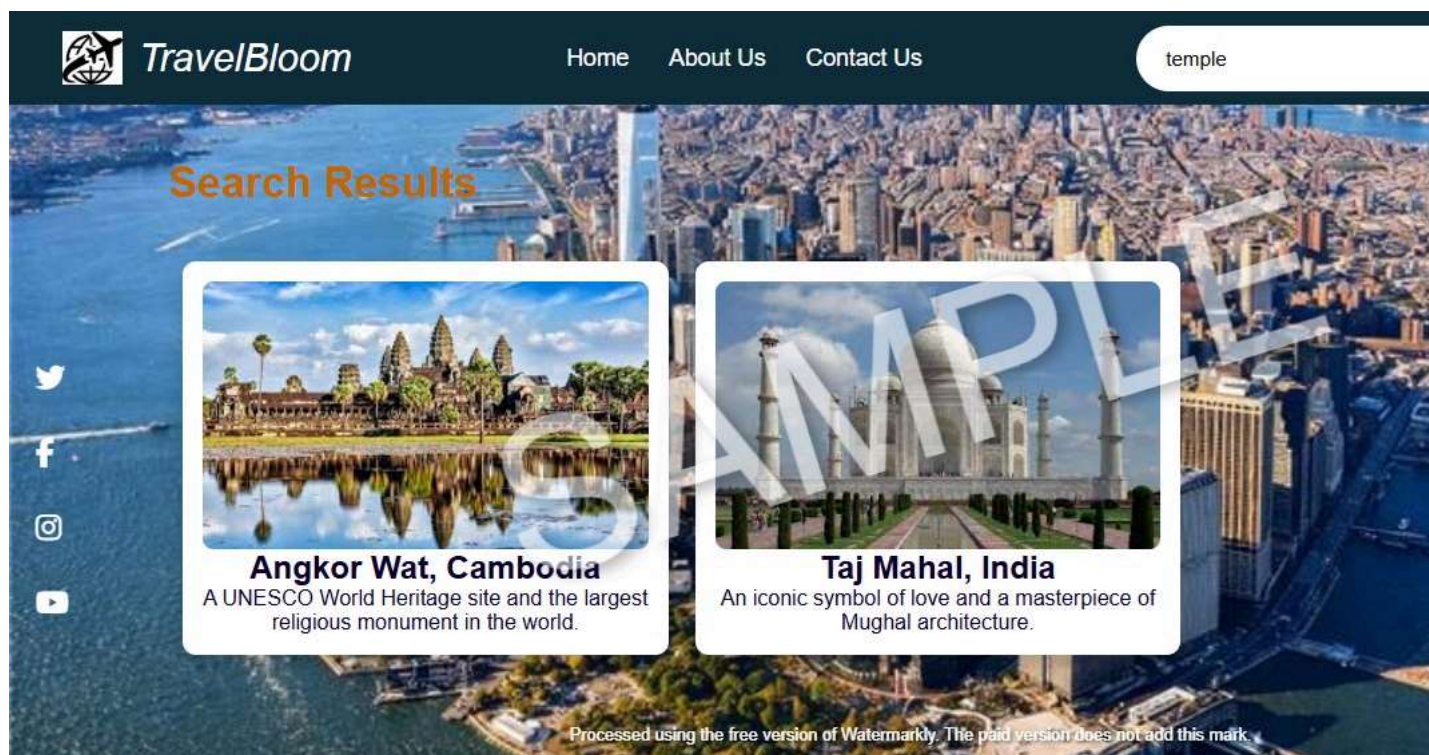
Assessment

Option 1: AI-Graded Submission and Evaluation

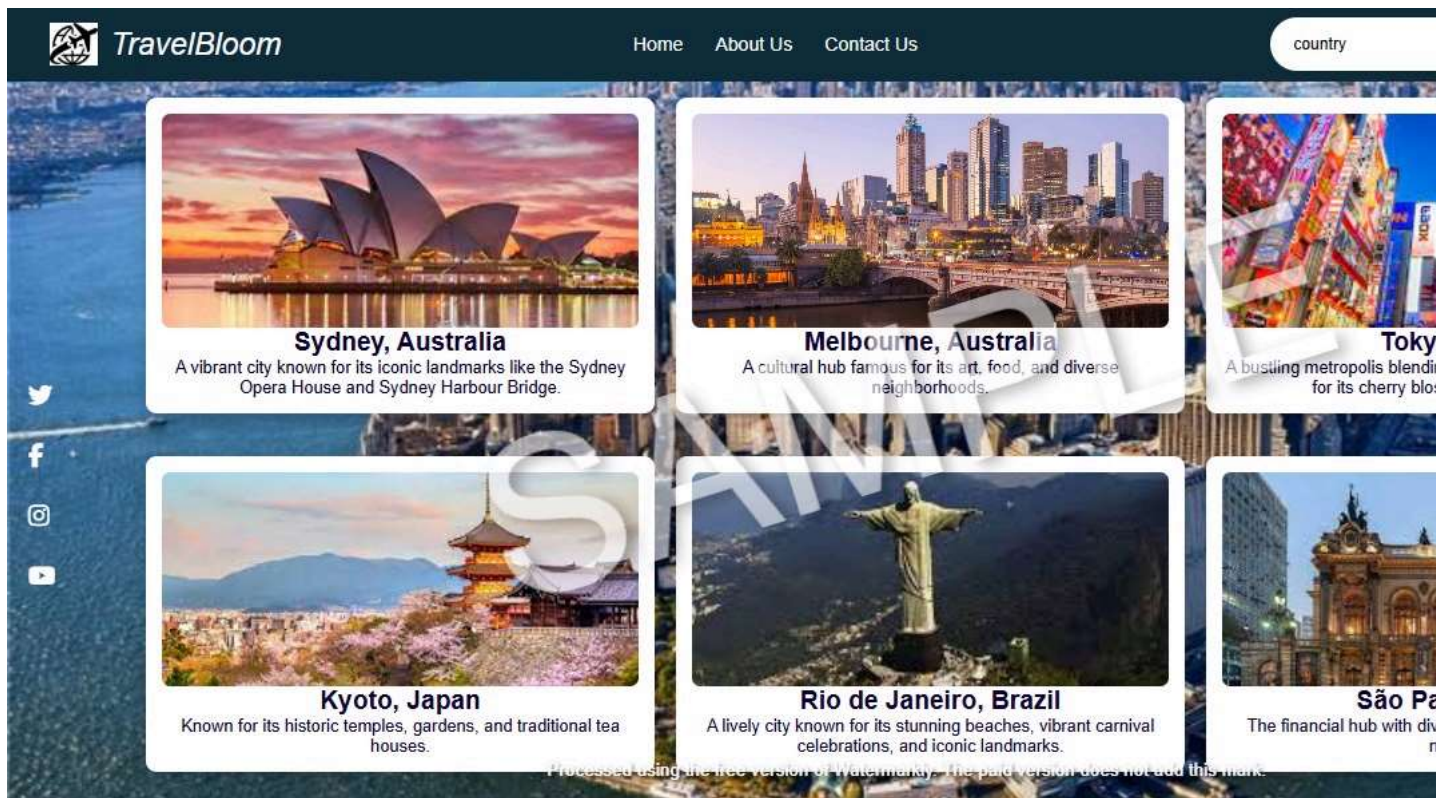
- Submission requires the public GitHub repository URL and the code snippets of the deployed Travel Recommendation project. Ensure all code is pushed to GitHub, the repository is up to date, and set to public.
- Ensure that your website is hosted on GitHub Pages and is publicly accessible and live.
- Submit a screenshot showing at least two beaches, saved as beach.jpeg or beach.png. A sample screenshot is provided for reference.



- Submit a screenshot showing at least two temples, saved as temple.jpeg or temple.png. .A sample screenshot is provided for reference.



- Submit a screenshot showing at least two countries, saved as country.jpeg or country.png. .A sample screenshot is provided for reference.



Option 2: Peer-Graded Submission and Evaluation

- Submission requires the public GitHub repository URL and the deployed URL of the Travel Recommendation project. Ensure all code is pushed to GitHub, the repository is up to date, and set to public.
- Ensure that your website is hosted on GitHub Pages and is publicly accessible and live.

Checklist for submission

Follow the checklist below to verify that your project meets all requirements before submission.

Submit your work through either Option 1: AI-Graded Submission and Evaluation or Option 2: Peer-Graded Submission and Evaluation, depending on the submission path you choose for project evaluation.

Checklist for Option 1: AI-Graded Submission and Evaluation

After completing the final project, you should have saved:

1. The valid public GitHub repository **URL** that contains the files related to the travel recommendation project.
2. The GitHub repository **URL** of your GitHub Pages deployed website contains the Home Page, including Home, About Us, and Contact Us sections, along with an introduction to the website.
3. The GitHub repository **URL** of the deployed website on GitHub Pages that includes the About Us page displaying the company's information and guidance or instructions for team members.
4. The GitHub repository **URL** of the deployed website on GitHub Pages that includes the Contact Us page with contact information and a form for users to reach you via email.
5. The code snippet of your deployed Travel Recommendation website clearly showing the navigation bar with these six elements: Home, About Us, Contact Us, Search bar input box, Search button, and Clear button, along with the visible live deployed URL.
6. The screenshot of your project showing that it recommends at least two beaches with images after typing any form of the word **beach** as the search term, saved as beach.jpeg or beach.png.
7. The screenshot of your project showing that it recommends at least two temples with images after typing any form of the word **temple** as the search term, saved as temple.jpeg or temple.png.
8. The screenshot of your project showing that it recommends at least two countries with images after typing any form of the word **country** as the search term, saved as country.jpeg or country.png.

Checklist for Option 2: Peer Review Evaluation

After completing the final project, you should have:

1. The valid public GitHub repository **URL** that contains the files related to the travel recommendation project.
2. The GitHub repository **URL** of the website deployed on GitHub Pages that contains:
 - The Home page, including an introduction to the website.
 - The About Us page displaying the company's information and guidance or instructions for team members.
 - The Contact Us page with contact information and a form allowing users to contact through email.

- The landing page with a navigation section showing all six menu items: Home, About Us, Contact Us, Search bar input box, Search button, and Clear button, along with the visible live deployed URL.
- At least two beaches with images after typing any form of the word **beach** as a search term.
- At least two temples with images after typing any form of the word **temple** as a search term.
- At least two countries with images after typing any form of the word **country** as a search term.

Summary

1. **HTML structure:** The HTML file establishes a structure comprising navigation links to **About Us** and **Contact Us** pages, and the **Home** page.
2. **JavaScript logic:** The embedded JavaScript code manages the user's query to display related information.
3. **Fetch data analysis capabilities:** The code includes functionalities to filter the user query keyword to fetch the data, which recommends travel options.
4. **Event handling:** Event listeners are set up to respond to user interactions with the navigation links and then apply event listeners for search and clear buttons.

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